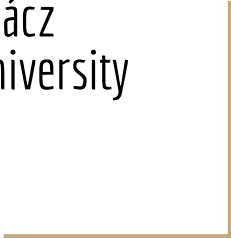




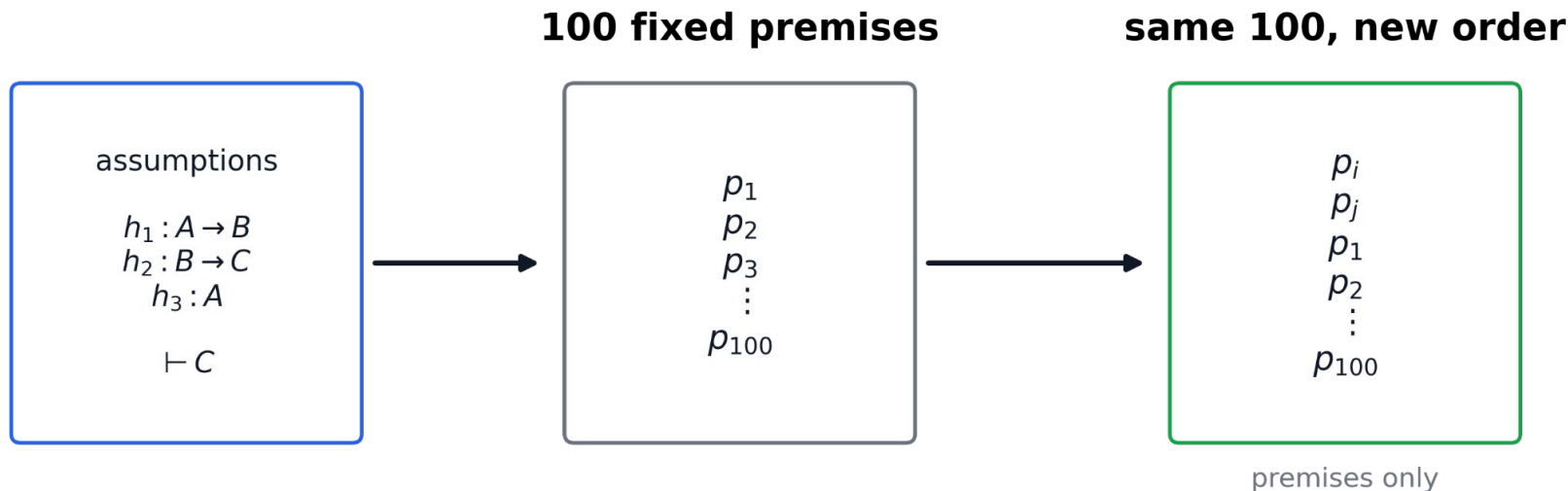
MITIGATING STATE DILUTION IN GRAPH NEURAL NETWORKS FOR PREMISE RERANKING VIA BLOCK ATTENTION RESIDUALS

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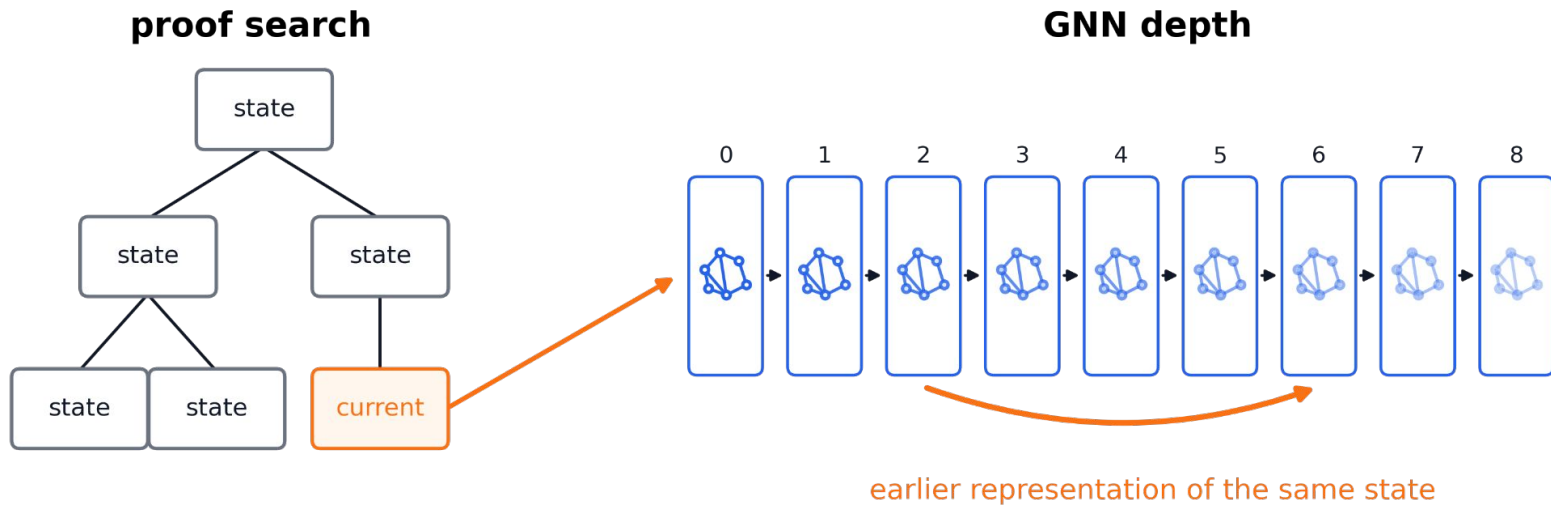
Implemented task: premise reranking

- Input: one current Lean proof state
- ReProver: fixed top-100 candidate premises
- Output: the same 100 premises, reordered
- No tactic generation or complete proof search



Proof-search depth GNN depth

- Proof search: verified state $\rightarrow$ verified state
- GNN depth: representation $\rightarrow$ representation
- Earlier source = earlier stage of the same current state

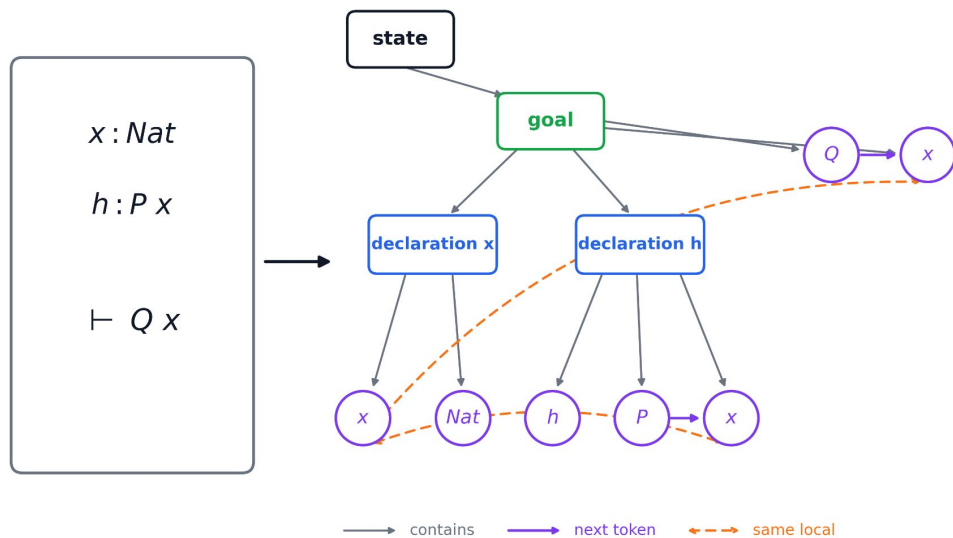


Can a deep GNN forget part of the proof state?

- Fixed-candidate Lean premise reranking
- Eight-layer relational GNN
- Selective access to earlier same-state representations

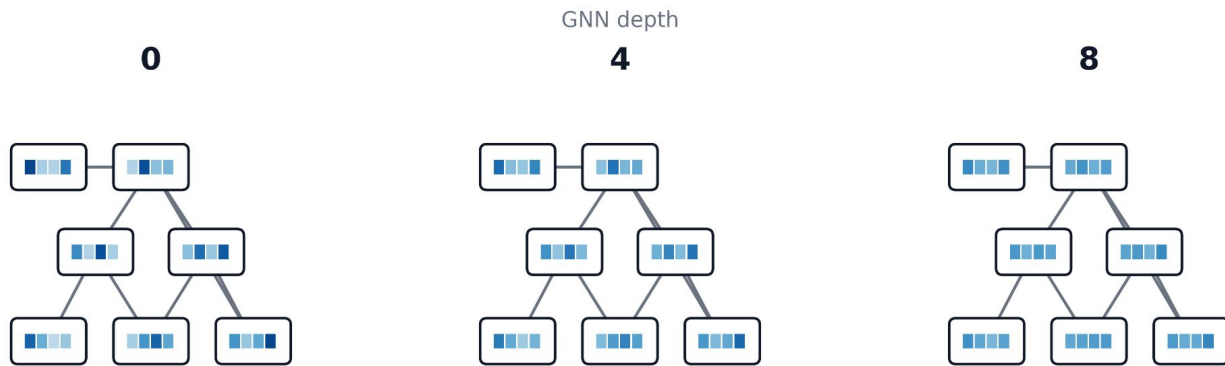
One Lean state becomes a typed graph

- Current goal + local assumptions
- Nodes: goals, declarations, tokens
- Typed relations preserve state structure



The graph stays fixed; representations change

- Graph topology remains fixed
- Hidden node features change across depth
- Repeated transformations may weaken input information

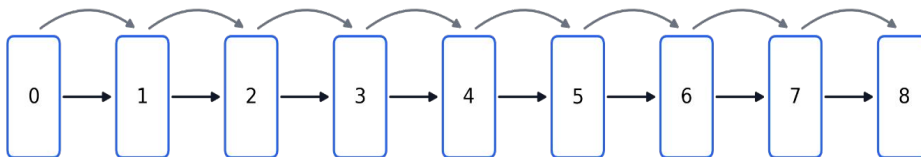


same nodes + edges | changing hidden vectors

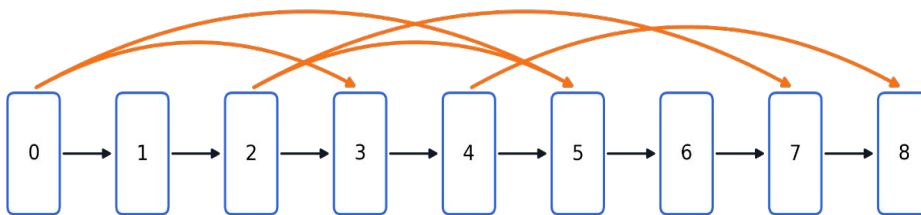
A controlled architectural comparison

- Same graph, node encoder, and eight RGCN transformations
- Same graph readout, candidates, scorer, and loss
- Same initialization and training protocol
- Only intended change: **depth connectivity**

Residual RGCN



BlockAttnRes



same graph
same encoder
same RGCN
same readout
same scorer

Mathematical Explanation

1) Historical Tensor Stacking

$$V \in \mathbb{R}^{S \times N \times 128}$$

2) Raw Attention Scoring

$$a_{s,n} = q_{\ell}^T K_{s,n}$$

3) Depth-Wise Softmax

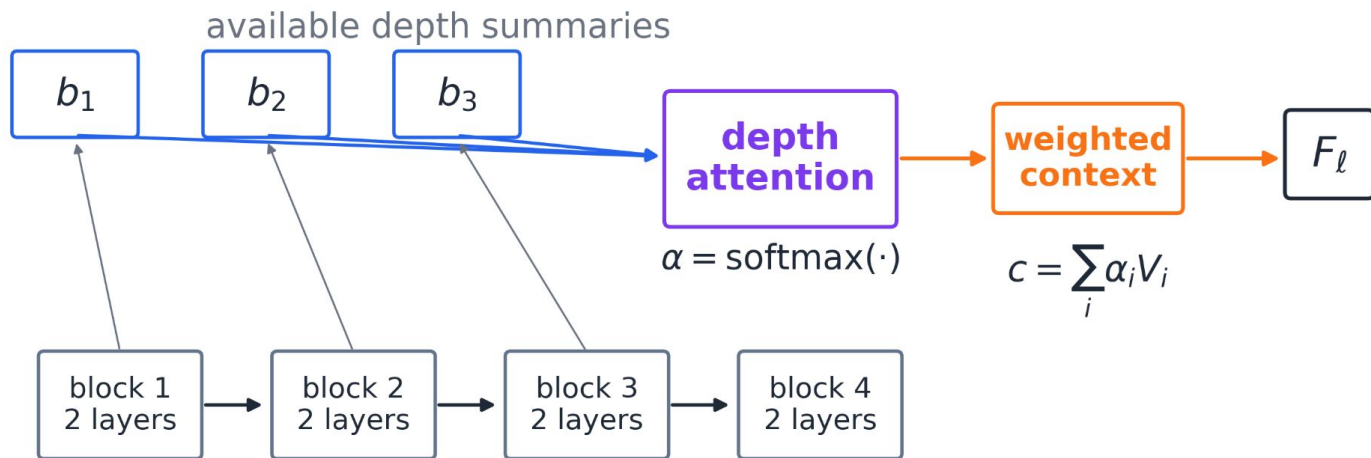
$$\alpha_{s,n} = \frac{\exp(a_{s,n})}{\sum_j \exp(a_{j,n})}$$

4) Context Retrieval

$$C_{\ell,n} = \sum_s \alpha_{s,n} V_{s,n}$$

BlockAttnRes gives each layer selective memory

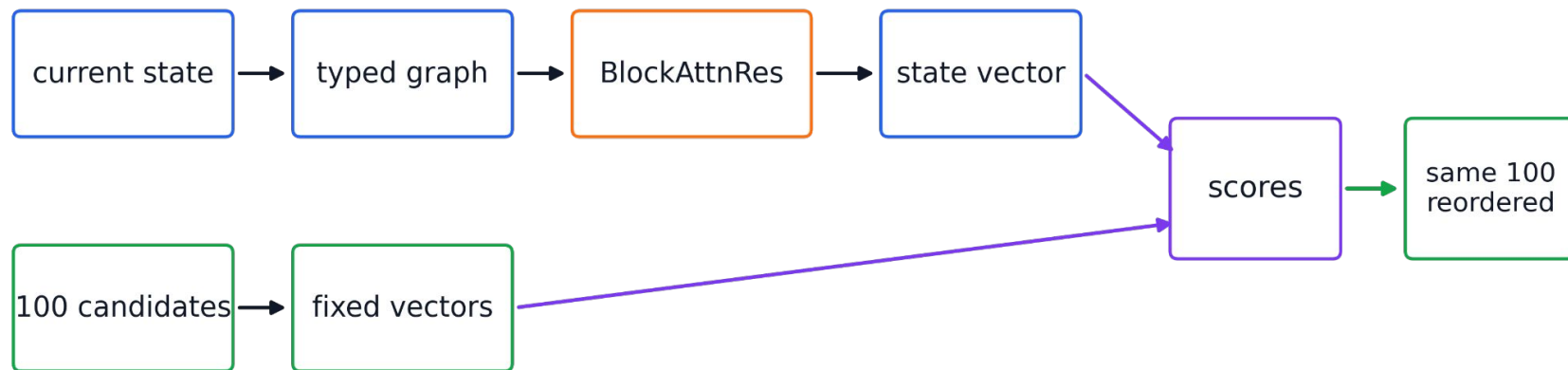
- Four blocks; two GNN layers per block
- Memory: initial, completed, and partial summaries
- Learned weights select among at most five depth sources



earlier block outputs are available for selection

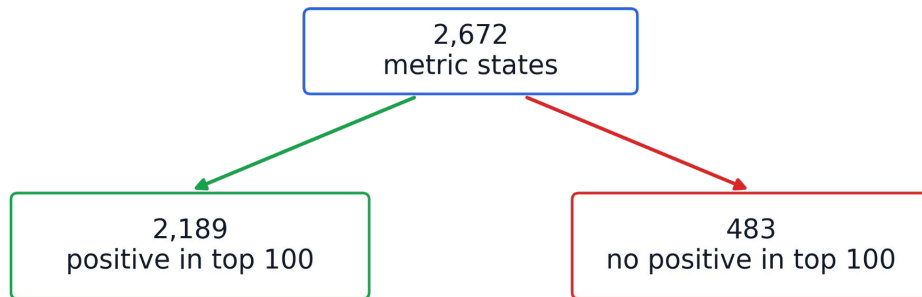
The state changes; the premise side stays fixed

- State branch: graph $\rightarrow$ 128-dimensional representation
- Premise branch: 100 frozen 1,472-dimensional vectors
- Shared scorer returns the same candidates in a new order



Evaluation: what counts as improvement?

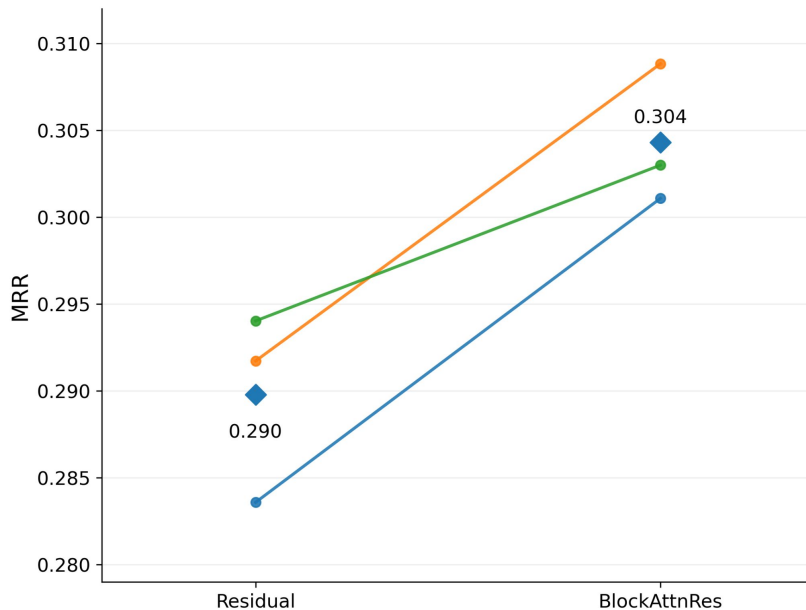
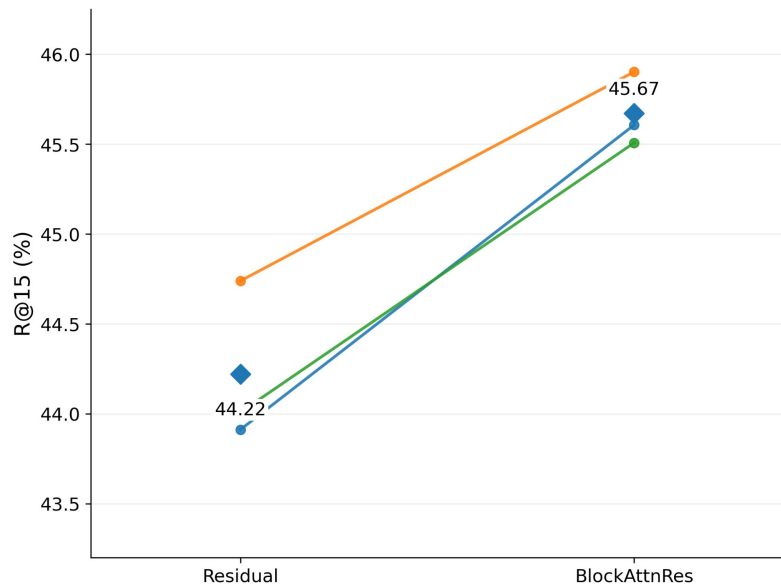
- R@15: relevant-premise recall within the first 15
- MRR: rank of the first relevant premise
- 2,672 metric states; 2,189 loss-eligible
- 483 states have no positive in the top 100



131 / 6,514 unresolved annotations

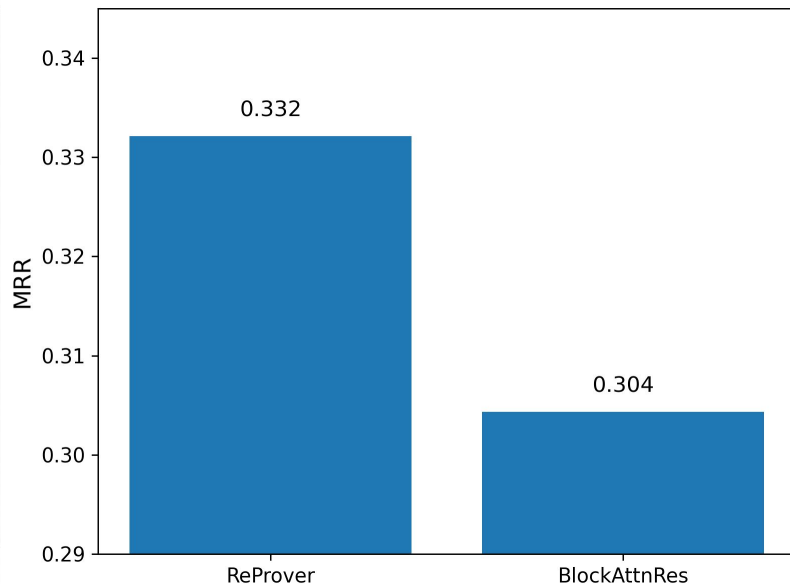
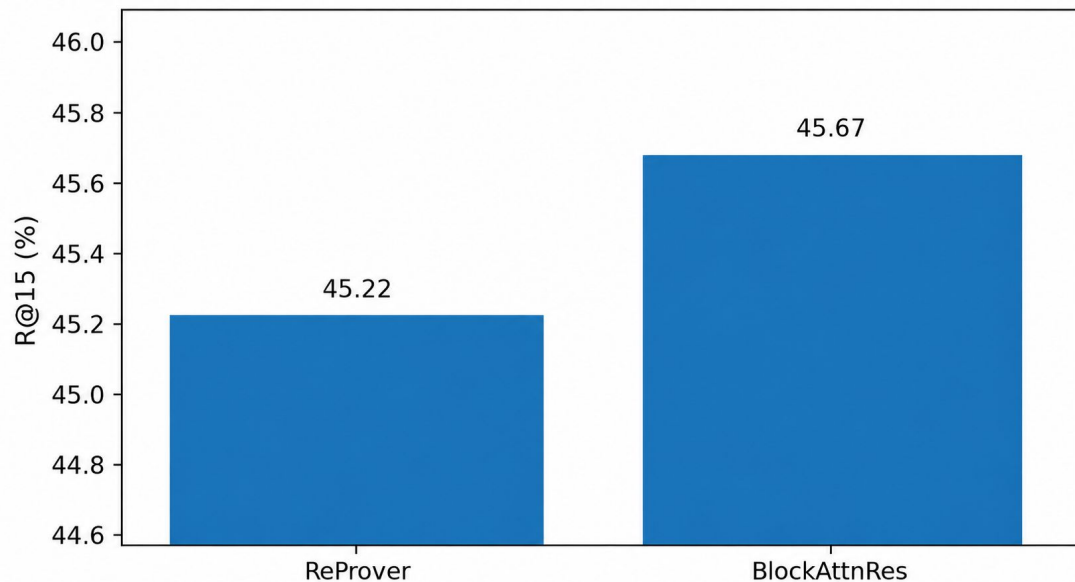
BlockAttnRes improves over the matched residual GNN

- R@15: 44.22% → 45.67%
- Gain: +1.45 percentage points
- MRR: 0.290 → 0.304
- Higher on all four metrics in all three paired seeds



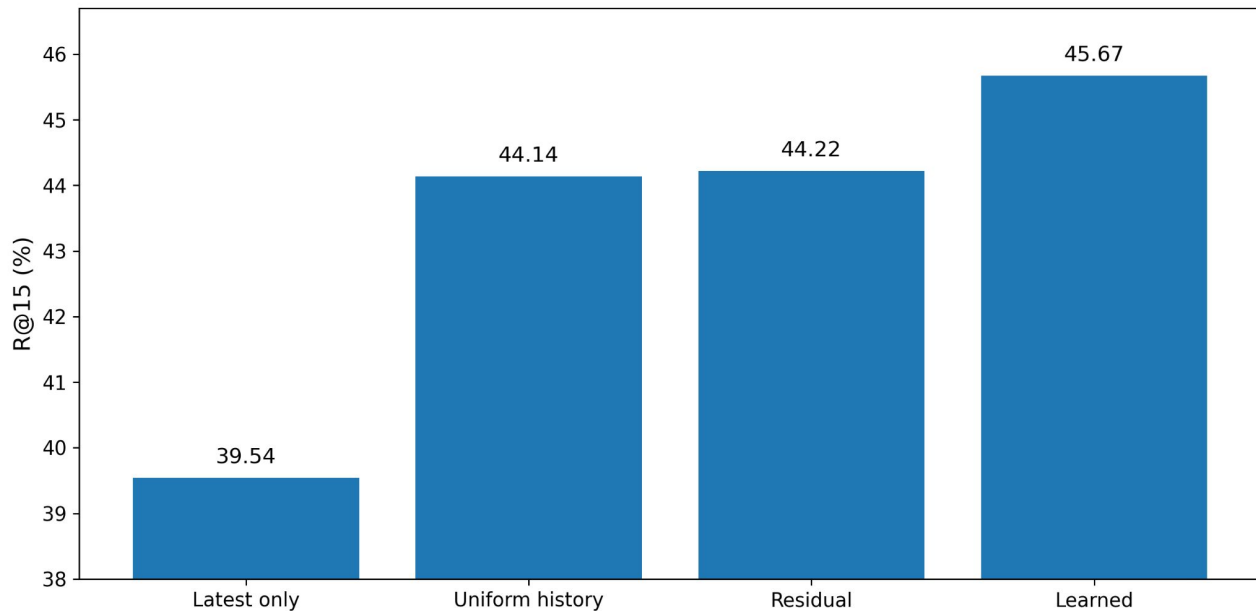
Against ReProver, the result is mixed

- BlockAttnRes: slightly higher R@15
- ReProver: higher R@1, R@10, and MRR
- More top-15 coverage; weaker very-early ranking



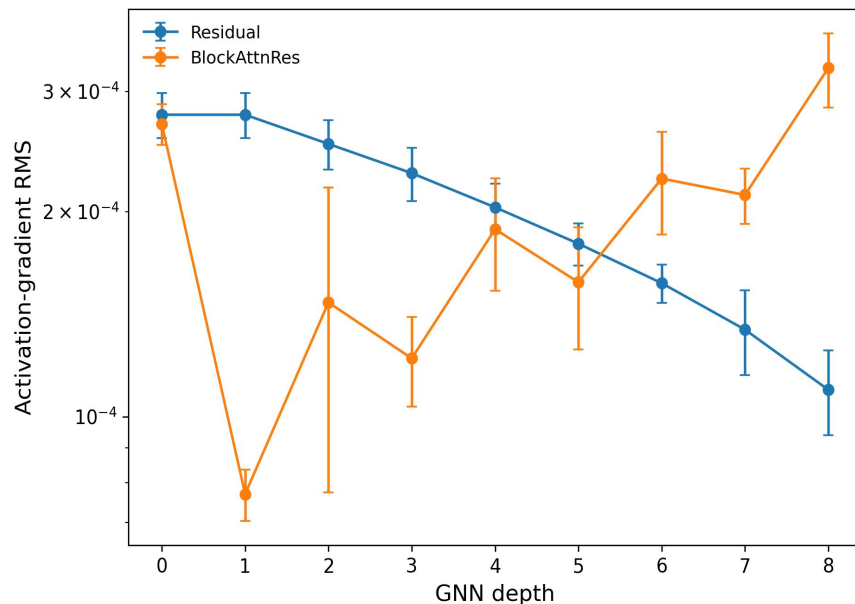
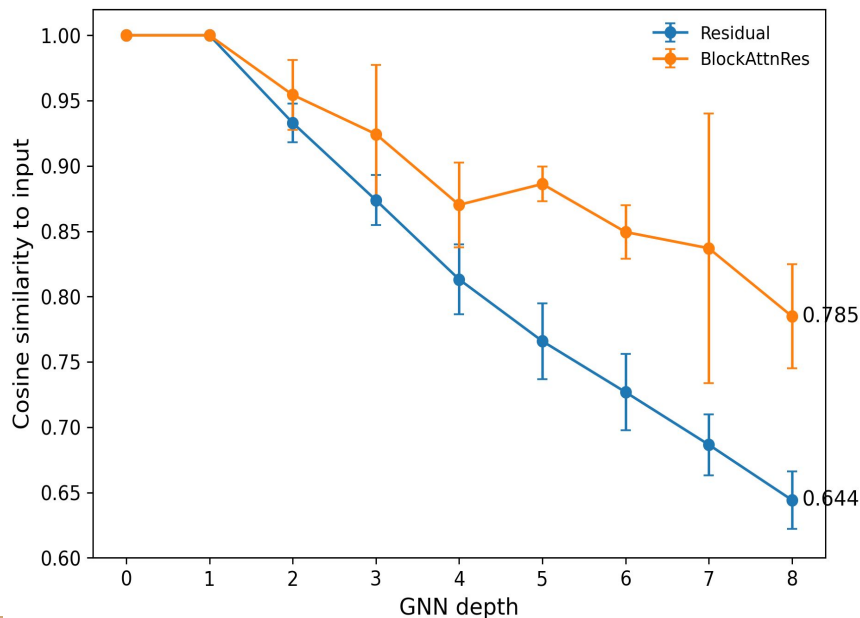
History access helps; learned selection creates the gain

- Latest-only processing causes a large performance drop
- Uniform history recovers most of the loss
- Learned selection exceeds both uniform history and residual RGCN



Internal measurements support the mechanism

- Directional similarity at depth 8: 0.785 vs 0.644
- Representation magnitude falls to 35% of the input
- Training signal shifts toward later layers
- 46.6% of final attention uses earlier sources

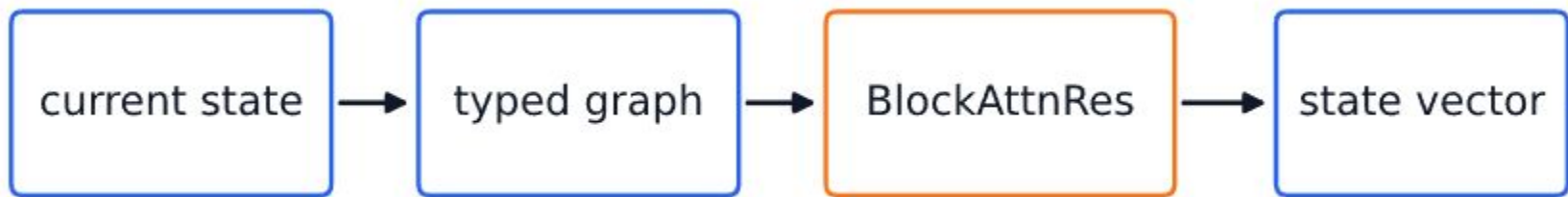


What this experiment does not establish

- Validation-only study; three paired seeds
- Premise reranking — not tactic generation or proof success
- Indirect diagnostics — not causal proof
- No Transformer, runtime, or total-memory comparison

Selective lookback improves fixed-candidate reranking

- BlockAttnRes improves over the matched residual GNN
- Ablations identify learned history selection as the key component
- Evidence is consistent with mitigating information loss across depth



Conclusions & Future Work

Conclusions

- **BlockAttnRes improves over the matched residual GNN**
- **Learned history selection is the key component**
- Evidence is consistent with **mitigating information loss across GNN depth**

Future work

- Evaluate on a **held-out test split** with more runs and deeper networks
- Benchmark **runtime and memory**
- Compare **broader encoder families**, including Transformers
- Integrate the reranker into **end-to-end Lean proof search**